

# Basic SQL

# SQL MURDER MYSTERY

SILVER

## FIRST CLUE...



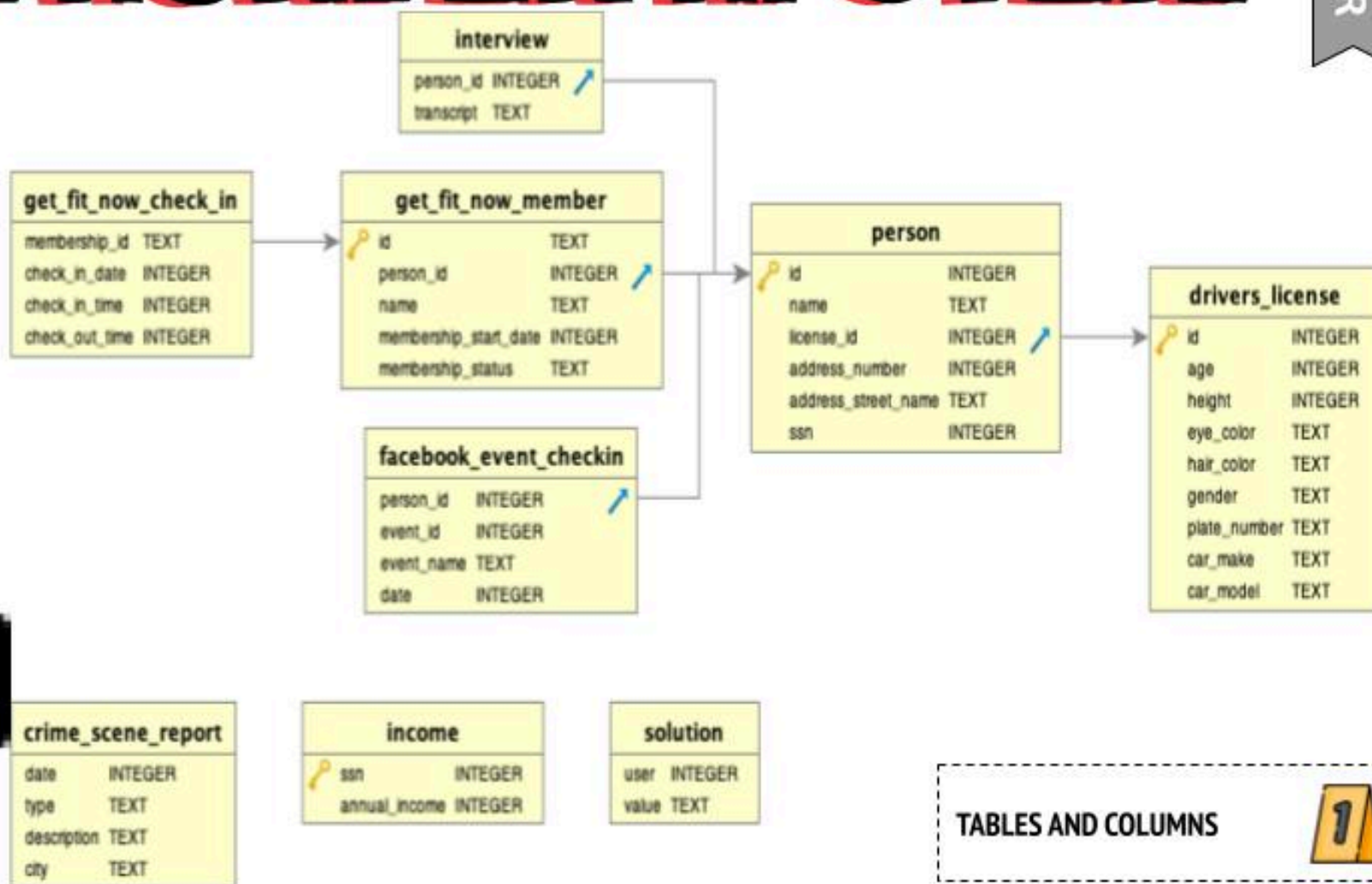
A crime has taken place and the detective needs your help. The detective gave you the crime scene report, but you somehow lost it. You vaguely remember that the crime was a **murder** that occurred sometime on **Jan.15, 2018** and that it took place in **SQL City**.

Start by retrieving the corresponding **crime scene report** from the police department's database.

*All the clues to this mystery are buried in a huge database, and you need to use SQL to navigate through this vast network of information. Your first step to solving the mystery is to retrieve the corresponding crime scene report from the police department's database.*

# SQL MURDER MYSTERY

SILVER



## To Write a Query:

**1** Which **table**?

**2** Which **columns**?

**3** Which **rows**?

# First clue

You vaguely remember that the crime was a murder that occurred sometime on Jan.15, 2018 and that it took place in SQL City. Start by retrieving the corresponding `crime_scene_report` from the police department's database.

- **1** A murder that occurred on Jan.15, 2018 in SQL City.
- **2** `crime_scene_report` table

**SELECT** columns **FROM** a table **WHERE** conditions  
are true

Table: crime\_scene\_report

Columns: all ( **\*** )

Rows: report type is murder

```
SELECT *  
FROM crime_scene_report  
WHERE type = 'murder';
```

# WHERE clause

Filters records based on specified conditions

- **Comparison operators** ( = , != , >= , ... )
- **Logical operators** ( NOT , AND , OR )
- **Membership operators** ( IN , NOT IN )

## **SELECT** all non-murder cases

```
-- using !=  
SELECT * FROM crime_scene_report WHERE type != 'murder';  
-- using not  
SELECT * FROM crime_scene_report WHERE NOT type = 'murder';
```



# Multiple matching criteria (AND, OR)

Table: crime\_scene\_report

Columns: all

Rows: a murder that occurred sometime on Jan.15, 2018 and that it took place in SQL City.

```
SELECT *  
FROM crime_scene_report  
WHERE type = 'murder'  
AND date = 20180115  
AND city = 'SQL City';
```

# BETWEEN

```
SELECT *  
FROM crime_scene_report  
WHERE type = 'murder'  
AND date BETWEEN 20180115 and 20180215;
```

```
SELECT *  
FROM crime_scene_report  
WHERE type = 'murder'  
AND date NOT BETWEEN 20180115 and 20180215;
```

# The Crime Report

```
SELECT *  
FROM crime_scene_report  
WHERE type = 'murder'  
AND date = 20180115  
AND city = 'SQL City';
```

Security footage shows that there were 2 witnesses. The first witness lives at the last house on "Northwestern Dr". The second witness, named Annabel, lives somewhere on "Franklin Ave".

- **1 Find the first witness who lives at the last house on "Northwestern Dr".**
- **2 Find the second witness named Annabel who lives on "Franklin Ave".**

# Matching strings

Table: person

Columns: all

Rows: lives at the last house on "Northwestern Dr"

```
SELECT *  
FROM person  
WHERE address_street_name = 'northWestern DR'
```

```
-- explicit case-insensitive matching  
SELECT *  
FROM person  
WHERE lower(address_street_name) = 'northwestern dr'
```

## ORDER BY address\_number (DESC ending order)

Table: person

Columns: all

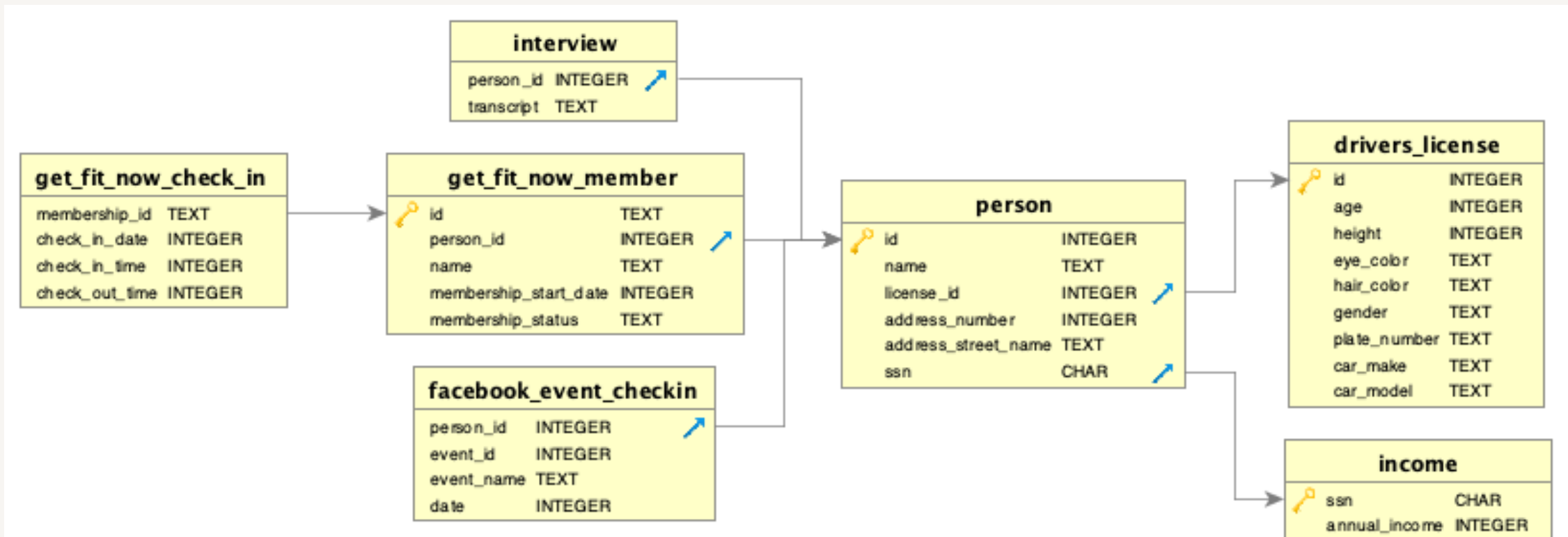
Rows: lives at *the last house* on "Northwestern Dr"

```
SELECT *  
FROM person  
WHERE address_street_name = 'Northwestern Dr'  
ORDER BY address_number DESC
```

| id    | name           | license_id | address_number | address_street_name | ssn       |
|-------|----------------|------------|----------------|---------------------|-----------|
| 14887 | Morty Schapiro | 118009     | 4919           | Northwestern Dr     | 111564949 |

# Find the interview of the first witness

| id    | name           | license_id | address_number | address_street_name | ssn       |
|-------|----------------|------------|----------------|---------------------|-----------|
| 14887 | Morty Schapiro | 118009     | 4919           | Northwestern Dr     | 111564949 |



# Primary Key & Foreign Key

**PK:** unique ID of each record in a table

**FK:** a column that references the PK of another table

**student** table

`id` is the  PK

| <code>id</code>  | <code>name</code> |
|---------------------------------------------------------------------------------------------------|-------------------|
| 2041                                                                                              | Tom               |
| 2204                                                                                              | John              |
| 2043                                                                                              | Alice             |

**mark** table

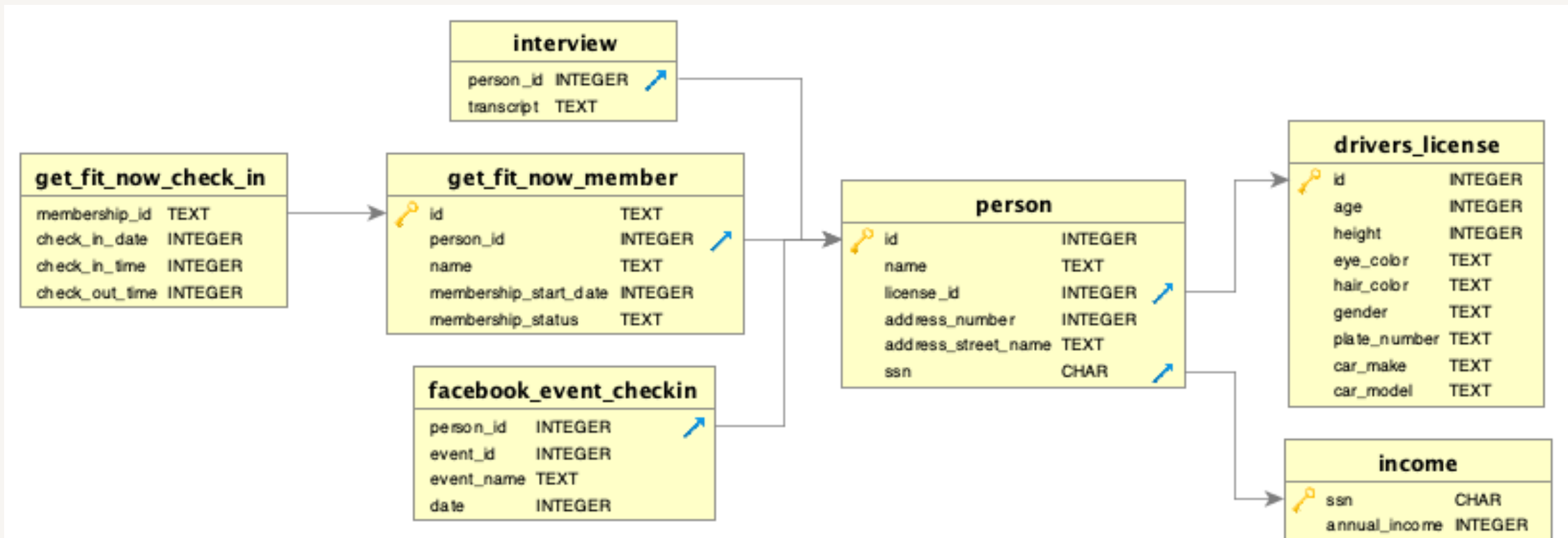
`student_id` is the  FK

| <code>student_id</code>  | <code>course</code> | <code>mark</code> |
|-------------------------------------------------------------------------------------------------------------|---------------------|-------------------|
| 2041                                                                                                        | SQL                 | 65                |
| 2041                                                                                                        | Python              | 73                |
| 2204                                                                                                        | SQL                 | 55                |
| 2043                                                                                                        | SQL                 | 90                |

# Connecting **interview** and **person** table?

- 🔗 **person\_id** in **interview** → 🔑 **id** in **person** table

| id    | name           | license_id | address_number | address_street_name | ssn       |
|-------|----------------|------------|----------------|---------------------|-----------|
| 14887 | Morty Schapiro | 118009     | 4919           | Northwestern Dr     | 111564949 |







## Who is Bruna Sumlin?

| id    | name         | license_id | address_number | address_street_name | ssn       |
|-------|--------------|------------|----------------|---------------------|-----------|
| 10325 | Bruna Sumlin | 340579     | 3808           | N Bartlett Ave      | 898899969 |

1. Find Bruna's interview.
2. Find Bruna's annual income.
3. Find Bruna's gym membership number.

# First witness's interview

```
SELECT * FROM interview WHERE person_id = 14887
```

I heard a gunshot and then saw a man run out. He had a "Get Fit Now Gym" bag. The membership number on the bag started with "48Z". Only gold members have those bags. The man got into a car with a plate that included "H42W".

- **1** Gold member whose id starts with "48Z" from `get_fit_now_member`
- **2** Driver whose license plate includes "H42W" from `drivers_license`

# Pattern matching **LIKE**

- **LIKE** operator for pattern matching in strings
- Wildcards:
  - **%**: Any number of characters
  - **\_**: Only one character

# Exact matching

```
-- exact matching  
SELECT * FROM get_fit_now_member WHERE id = '48Z'  
SELECT * FROM get_fit_now_member WHERE id LIKE '48Z'
```

# Pattern matching

## Starts with:

```
-- followed by any number of characters
SELECT * FROM get_fit_now_member WHERE id LIKE '48Z%'
-- followed by only one character
SELECT * FROM get_fit_now_member WHERE id LIKE '48Z_'
```

## Ends with:

```
-- any number of preceeding characters
SELECT * FROM get_fit_now_member WHERE id LIKE '%48Z'
-- only one preceeding character
SELECT * FROM get_fit_now_member WHERE id LIKE '_48Z'
```

## Includes:

```
SELECT * FROM get_fit_now_member WHERE id LIKE '%48Z%'
```

**1 Gold member whose id starts with "48Z" from** `get_fit_now_member`

```
SELECT * FROM get_fit_now_member  
WHERE id LIKE '48Z%'  
AND membership_status='gold'
```

| id    | person_id | name          | membership_start_date | membership_status |
|-------|-----------|---------------|-----------------------|-------------------|
| 48Z55 | 67318     | Jeremy Bowers | 20160101              | gold              |
| 48Z7A | 28819     | Joe Germuska  | 20160305              | gold              |

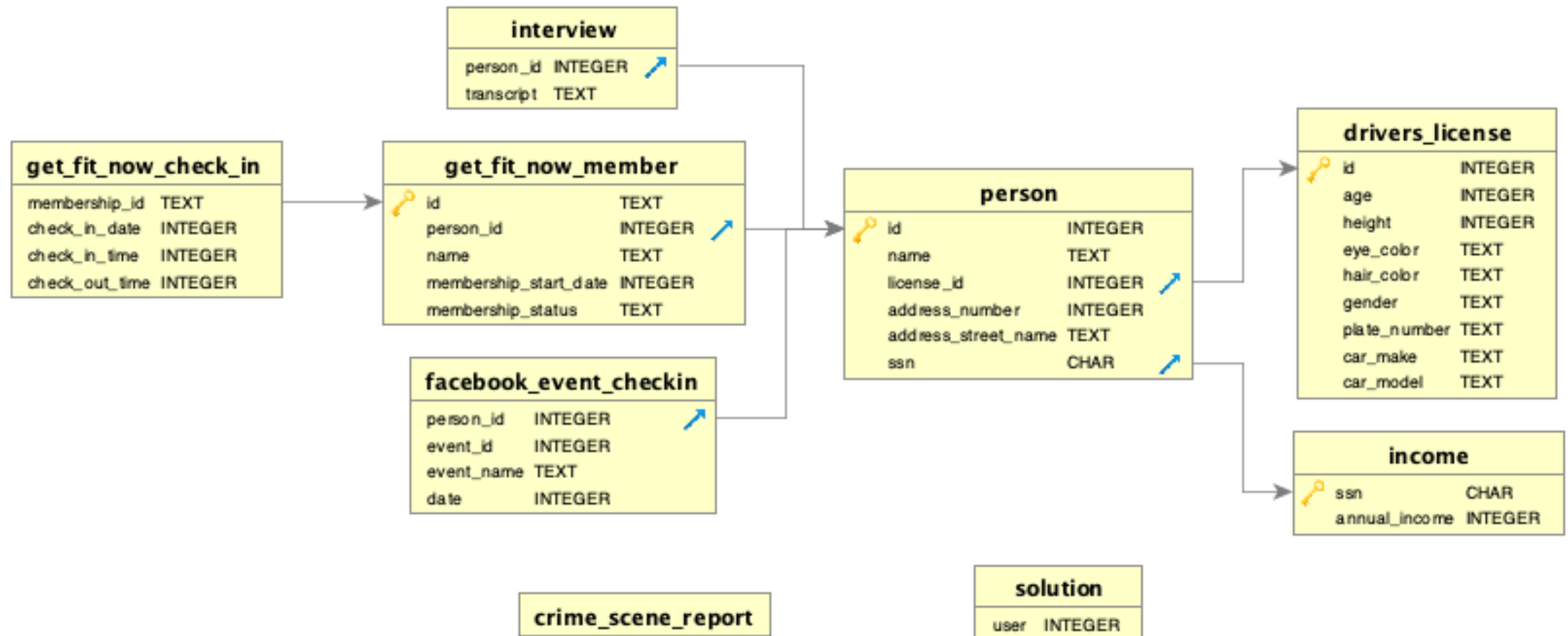
## 2 Driver whose license plate includes "H42W" from `drivers_license`

```
SELECT * FROM drivers_license  
WHERE plate_number LIKE '%H42W%'
```

| id     | age | height | eye_color | hair_color | gender | plate_number | car_make  | car_model |
|--------|-----|--------|-----------|------------|--------|--------------|-----------|-----------|
| 183779 | 21  | 65     | blue      | blonde     | female | H42W0X       | Toyota    | Prius     |
| 423327 | 30  | 70     | brown     | brown      | male   | 0H42W2       | Chevrolet | Spark LS  |
| 664760 | 21  | 71     | black     | black      | male   | 4H42WR       | Nissan    | Altima    |

# Connecting `get_fit_now_member` to `drivers_license`?

-  `person_id` in `get_fit_now_member` →  `id` in `person`
-  `license_id` in `person` → `id` in  `drivers_license`





# Cross-checking two tables via the `person` table

Using logical operators ( `AND` , `OR` , `NOT` )

```
SELECT * FROM person
-- from clue1 (get_fit_now_member)
WHERE (id = 67318 OR id = 28819)
-- from clue2 (drivers_license)
AND (license_id = 183779 OR license_id = 423327 OR license_id = 664760)
```

Using membership operators ( `IN` , `NOT IN` )

```
SELECT * FROM person
-- from clue1 (get_fit_now_member)
WHERE id IN (67318, 28819)
-- from clue2 (drivers_license)
AND license_id IN (183779, 423327, 664760)
```



## Find the second witness

```
SELECT *  
FROM crime_scene_report  
WHERE type = 'murder'  
AND date = 20180115  
AND city = 'SQL City';
```

Security footage shows that there were 2 witnesses. The first witness lives at the last house on "Northwestern Dr". The second witness, named Annabel, lives somewhere on "Franklin Ave".

- **1** ~~Find the first witness who lives at the last house on "Northwestern Dr".~~
- **2** Find the second witness named Annabel who lives on "Franklin Ave".

# To Write a Query:

**1** Which **table**?

**2** Which **columns**?

**3** Which **rows**?

**2** Find the second witness named Annabel who lives on "Franklin Ave".

1. What's the person\_id of the second witness?
2. What's on her interview?
3. What's her gym membership\_id?
4. Figure out what time she was at the gym
5. Who else was at the gym during that time?
6. Find out their names.



## What's the `person_id` of the second witness?

The second witness, named Annabel, lives somewhere on "Franklin Ave".



## What's on her interview?

Hint: Use the `person_id` to find her interview.



## What's her gym membership\_id?

Hint: Use the person\_id to find her membership details.



**Figure out what time she was at the gym**

 Who else was at the gym during the time she was there?

Hint: Anyone who left **before** she arrived or arrived **after** she left should be excluded.





## Find out their names

Hint: Use the membership IDs

# Wrap up

- `SELECT` columns `FROM` a table
- `WHERE` conditions are true
  - `=`, `!=`, `>=`, ... comparison operators
  - `NOT`, `AND`, `OR` logical operators
  - `LIKE` pattern matching
  - `IN` membership operator
  - `BETWEEN` range operator
- `LIMIT` the number of records returned
- `ORDER BY` columns